Scheme of Valuation – ROBOTICS (MG4DSEECT204)

Part A – MCQ (16 × 1 = 16 Marks)

- Correct option: **1 mark**
- Wrong answer / no answer: **0 mark**

c) Camera

c) Number of ways robot can move or position

b) Binary full-on or full-off based on setpoint

a) Teaching programming

b) Minimal maintenance and resilience to hazards

a) Learning and decision-making

a) Planning: path generation; Tracking: follow via control

b) Ecosystem packages community

d) Speed and accuracy

b) Perform repetitive manufacturing tasks

d) Transmitter and receiver

a) Linear: direct thrust no rotary; DC: continuous rotation

b) Environmental conditions

b) Planar pick-place tasks

b) Hardware abstraction portability

c) Stress safety and integration checks

Part B – Short Answer Questions (Any 6 × 5 = 30 Marks)

Q17. Robot Classification (5 Marks)

- Definition of robot classification – 2 marks
- Basis of classification (mobility, structure, control, application, DOF) – 2 marks
- Examples of robot categories – 1 mark

Q18. Path Planning in Robotics (5 Marks)

- Definition – 2 marks
- Collision-free path concept – 1 mark
- Consideration of obstacles/safety/efficiency – 2 marks

Q19. Conceptual Design of Mobile Robot (5 Marks)

- Requirements and payload identification – 1 mark
- Locomotion selection – 1 mark
- Sensor selection – 1 mark
- Controller and power source – 1 mark
- Prototype/testing/refinement – 1 mark

Q20. Sensors and Control for Safe Human–Robot Interaction (5 Marks)

- Role of sensors – 2 marks
- Examples of sensors – 1 mark
- Role of control system – 1 mark
- Safety measures (slowdown/stop/collision avoidance) – 1 mark

Q21. Robotic Locomotion (5 Marks)

- Definition – 2 marks
- Types of locomotion – 2 marks
- Suitable examples – 1 mark

Q22. Principles of Motion in Robotic Arm (5 Marks)

- Motion concepts (displacement, velocity, acceleration) – 2 marks
- Joint movement/coordination – 2 marks
- Practical example – 1 mark

Q23. Feedback Control of Robot Motor (5 Marks)

- Definition of feedback control – 2 marks
- Sensor measurement of speed – 1 mark
- Error comparison and correction – 1 mark
- Maintaining constant speed – 1 mark

Q24. Humidity Sensors in Weather Monitoring (5 Marks)

- Definition/function – 2 marks
- Importance in weather prediction – 2 marks
- Examples/applications – 1 mark

Part C – Essay Questions (Any 2 × 12 = 24 Marks)

Q25. Robotics in Modern Society (12 Marks)

Component	Marks
Introduction	2
Roles/applications	4
Impact on daily life	3
Challenges	2
Conclusion	1

Q26. Tracked and Humanoid Robots (12 Marks)

Component	Marks
Tracked robots (advantages, limitations, applications)	5
Humanoid robots (advantages, limitations, applications)	5
Comparison and conclusion	2

Q27. Sensors, Intelligence and Autonomy (12 Marks)

Component	Marks
Introduction	1
Types of sensors	3
Contribution to intelligence	3
Contribution to autonomy	3
Examples/conclusion	2

Q28. Space Robots vs Underwater Robots (12 Marks)

Component	Marks
Environment comparison	2
Design requirements	2
Sensors	2
Communication and power systems	2
Operational challenges	2
Applications/conclusion	2